

1 Semi-implicit

$$\begin{pmatrix} u_1 \\ h_1 \end{pmatrix} = \begin{pmatrix} 1 - (1 - \alpha) cFr\Delta tk & -(1 - \alpha) c\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} -\alpha cFr\Delta tk & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix}$$

$$\begin{aligned} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ -\alpha c\Delta tk & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} \\ &+ \begin{pmatrix} 0 & 0 \\ -(1 - \alpha) c\Delta tk & -(1 - \alpha) cFr\Delta tk \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} \\ &+ \begin{pmatrix} 0 & 0 \\ 0 & -\alpha cFr\Delta tk \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix} \end{aligned}$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c\Delta t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix}$$

this implies

$$\begin{aligned} \begin{pmatrix} u^i \\ h^i \end{pmatrix} &= \begin{pmatrix} 1 & -\alpha c\Delta t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix}^{-1} \\ &\left[\begin{pmatrix} 1 & 0 \\ -\alpha c\Delta tk & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ -(1 - \alpha) c\Delta tk & -(1 - \alpha) cFr\Delta tk \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & -\alpha cFr\Delta tk \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix} \right] \end{aligned}$$

$$\begin{aligned} \begin{pmatrix} u^i \\ h^i \end{pmatrix} &= \begin{pmatrix} 1 & -\alpha c\Delta t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix}^{-1} \\ &\left[\begin{pmatrix} 1 & 0 \\ -\alpha c\Delta tk & 1 \end{pmatrix} \left[\begin{pmatrix} 1 - (1 - \alpha) cFr\Delta tk & -(1 - \alpha) c\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} -\alpha cFr\Delta tk & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix} \right] + \end{aligned}$$

$$\begin{aligned} \begin{pmatrix} u^i \\ h^i \end{pmatrix} &= \begin{pmatrix} 1 & -\alpha c\Delta t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix}^{-1} \\ &\left[\begin{pmatrix} 1 - (1 - \alpha) cFr\Delta tk & -(1 - \alpha) c\Delta tk \\ -\alpha c\Delta tk(1 - (1 - \alpha) cFr\Delta tk) - (1 - \alpha) c\Delta tk & 1 - (1 - \alpha) cFr\Delta tk + \alpha(1 - \alpha) c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} \right] \end{aligned}$$

2 Segregated Implicit Advection, Implicit Gravity

$$\begin{pmatrix} 1 + \alpha cFr\Delta tk & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} = \begin{pmatrix} 1 - (1 - \alpha) cFr\Delta tk & -(1 - \alpha) c\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & -\alpha c\Delta tk \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix}$$

$$\begin{pmatrix} u_2 \\ h_2 \end{pmatrix} = \begin{pmatrix} 1 - (1 - \alpha) cFr\Delta tk & -(1 - \alpha) c\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} -\alpha cFr\Delta tk & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 + \alpha cFr\Delta tk - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u_3 \\ h_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -(1 - \alpha) c\Delta tk & 1 - (1 - \alpha) cFr\Delta tk \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ -\alpha c\Delta tk & 0 \end{pmatrix}$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_3 \\ h_3 \end{pmatrix}$$

this implies

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \alpha cFr\Delta tk - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 1 & 0 \\ -(1 - \alpha) c\Delta tk & 1 - (1 - \alpha) cFr\Delta tk \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} \right]$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \alpha cFr\Delta tk - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 1 & 0 \\ -(1 - \alpha) c\Delta tk & 1 - (1 - \alpha) cFr\Delta tk \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ -\alpha c\Delta tk & 0 \end{pmatrix} \left[\begin{pmatrix} 1 - (1 - \alpha) cFr\Delta tk & -(1 - \alpha) c\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} \right] \right]$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \alpha cFr\Delta tk - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 1 & 0 \\ \alpha(1 - \alpha) c^2 Fr\Delta t^2 k^2 - c\Delta tk & 1 - (1 - \alpha) cFr\Delta tk + \alpha(1 - \alpha) c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 \\ \alpha^2 c^2 Fr\Delta t^2 k^2 \end{pmatrix} \right]$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c\Delta tk \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \alpha cFr\Delta tk - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 1 & 0 \\ \alpha(1 - \alpha) c^2 Fr\Delta t^2 k^2 - c\Delta tk & 1 - (1 - \alpha) cFr\Delta tk + \alpha(1 - \alpha) c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 \\ \alpha^2 c^2 Fr\Delta t^2 k^2 \end{pmatrix} \right]$$

3 Operator Split Implicit Advection (both) and Implicit Gravity

$$\begin{pmatrix} 1 + \alpha cFr\Delta tk & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} = \begin{pmatrix} 1 - (1 - \alpha) cFr\Delta tk & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 + \alpha c Fr \Delta t k - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ -\alpha c \Delta t k & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -(1-\alpha) c \Delta t k & 1 - (1-\alpha) c Fr \Delta t k + (1-\alpha)^2 c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c \Delta t k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix} + \begin{pmatrix} 0 & -(1-\alpha) c \Delta t k \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

which implies

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c \Delta t k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \alpha c Fr \Delta t k - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 0 & 0 \\ -\alpha c \Delta t k & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -(1-\alpha) c \Delta t k & 1 - (1-\alpha) c Fr \Delta t k + (1-\alpha)^2 c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} \right] + \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c \Delta t k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 + \alpha c Fr \Delta t k - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix}^{-1} \left[\begin{pmatrix} 0 & 0 \\ -\alpha c \Delta t k & 0 \end{pmatrix} \begin{pmatrix} 1 + \alpha c Fr \Delta t k & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 - (1-\alpha) c Fr \Delta t k & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -(1-\alpha) c \Delta t k & 1 - (1-\alpha) c Fr \Delta t k + (1-\alpha)^2 c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} \right] + \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

no dependence on previous iteration in this case - so doesn't improve with more iterations

4 Operator Split Implicit Momentum Advection, Explicit Mass Advection and Implicit Gravity

$$\begin{pmatrix} 1 + \alpha c Fr \Delta t k & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} = \begin{pmatrix} 1 - (1-\alpha) c Fr \Delta t k & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ -\alpha c \Delta t k & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -(1-\alpha) c \Delta t k & 1 - (1-\alpha) c Fr \Delta t k + (1-\alpha)^2 c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c \Delta t k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_2 \\ h_2 \end{pmatrix} + \begin{pmatrix} 0 & -(1-\alpha) c \Delta t k \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$$

which implies

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c \Delta t k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix}^{-1} \\ \left[\begin{pmatrix} 0 & 0 \\ -\alpha c \Delta t k & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ h_1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -(1-\alpha) c \Delta t k & 1 - (1-\alpha) c F r \Delta t k + (1-\alpha)^2 c^2 \Delta t^2 k^2 \end{pmatrix} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right]$$

$$\begin{pmatrix} u^i \\ h^i \end{pmatrix} = \begin{pmatrix} 1 & -\alpha c \Delta t k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \alpha^2 c^2 \Delta t^2 k^2 \end{pmatrix}^{-1} \\ \left[\left\{ \begin{pmatrix} 0 & 0 \\ -\alpha c \Delta t k & 0 \end{pmatrix} \begin{pmatrix} 1 + \alpha c F r \Delta t k & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 - (1-\alpha) c F r \Delta t k & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ -(1-\alpha) c \Delta t k & 1 - (1-\alpha) c F r \Delta t k + (1-\alpha)^2 c^2 \Delta t^2 k^2 \end{pmatrix} \right\} \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right]$$

now there is dependence on previous iterations